

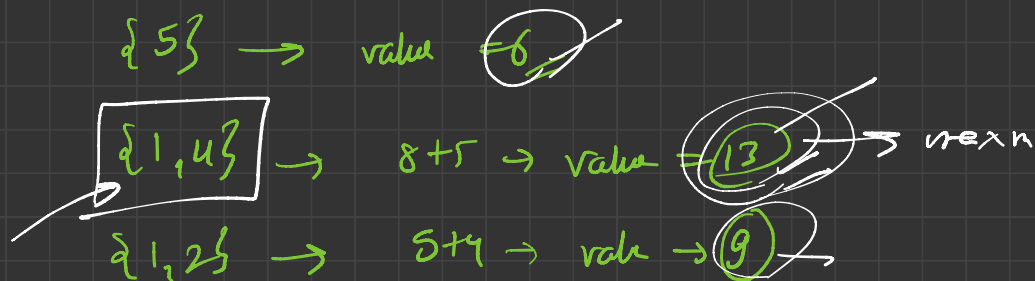
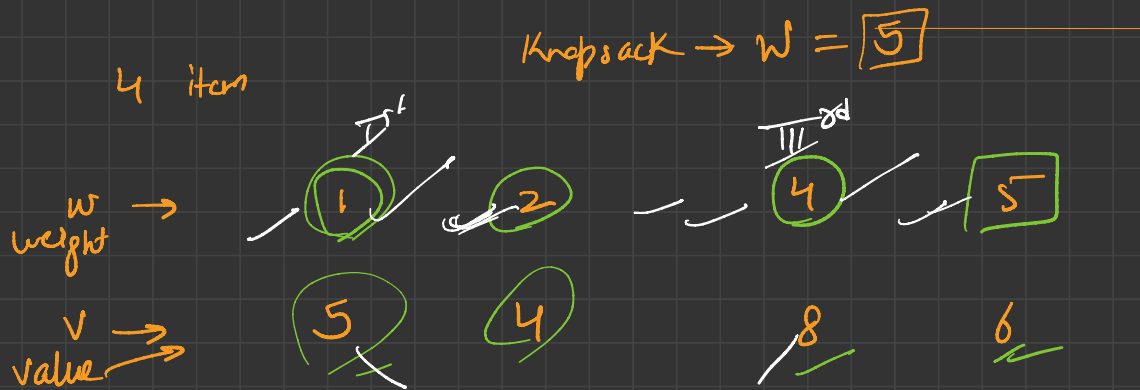

Knapsack Problem

Thief $\rightarrow$ Knapsack $\rightarrow$ can carry W $\leftarrow$

n item

1	2	3	4	5	...	n^{th}
w_1	w_2	w_3	w_4			w_n
v_1	v_2	v_3	v_4			v_n

value maxi



$\{1, 2, 3, 4\} \rightarrow$ $\{1\}$ $\{1, 2\}$ $\{1, 3\}$ $\{1, 4\}$ $\{2, 3\}$ $\{2, 4\}$

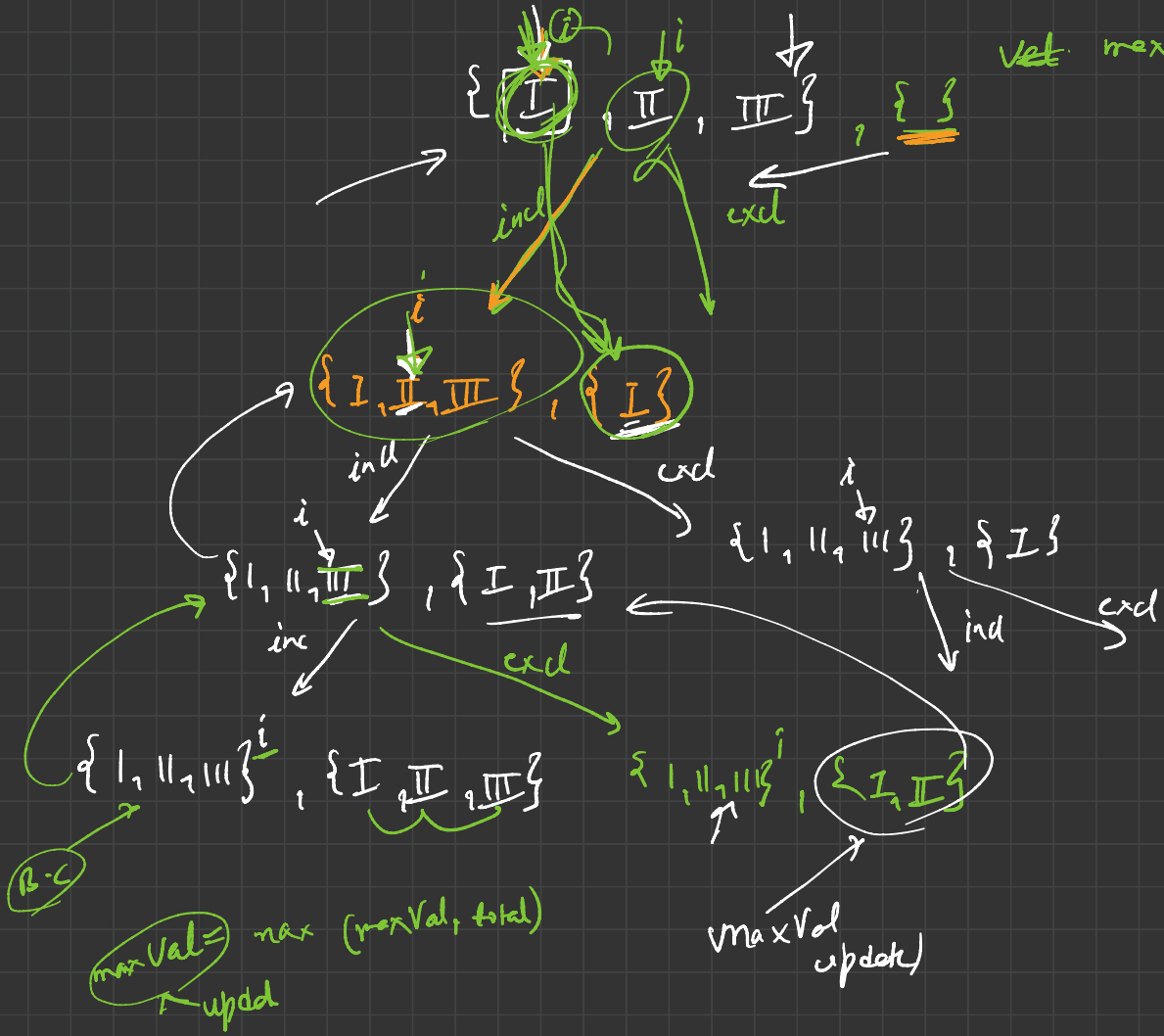
Approach → Brute force

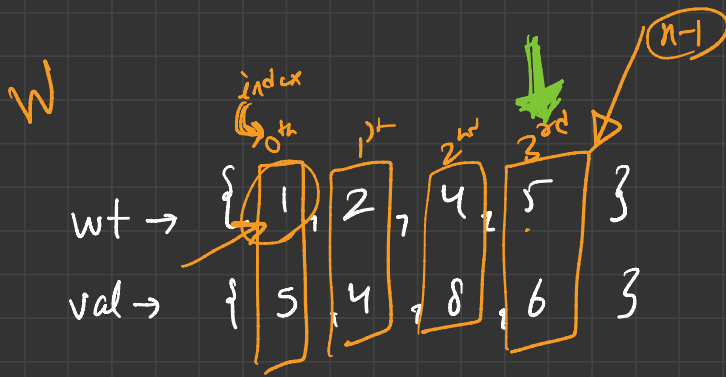
3 items

A, B, C

- ① $\{A\}_{val}$ ② $\{A, B\}$ ③ $\{A, B, C\}$
- ④ $\{B\}_{val}$ ⑤ $\{A, C\}$ ⑥ $\{\}$
- ⑦ $\{C\}_{val}$ ⑧ $\{B, C\}$

Let $maxVal = INT_MIN$





B.C

1 item

W

if (wt[0] ≤ maxWeight)

return val[0]

else

return 0;

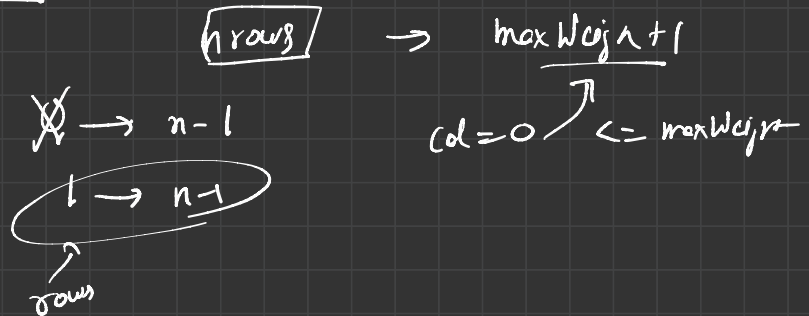
① Recursive + Memoization (Top-Down)

Step ① $\rightarrow$ dp array $\rightarrow (-1)$

Step 2 $\rightarrow$ Rec call $\rightarrow$ dp $\rightarrow$ arr store
return

Step 3 $\rightarrow$ B.C $\rightarrow$ if dp array contains arr
then return

② Bottom-up Approach (Tabulation)



0

row $\rightarrow n$

col $\rightarrow \maxWeight + 1$

~~0~~ $\rightarrow n-1$

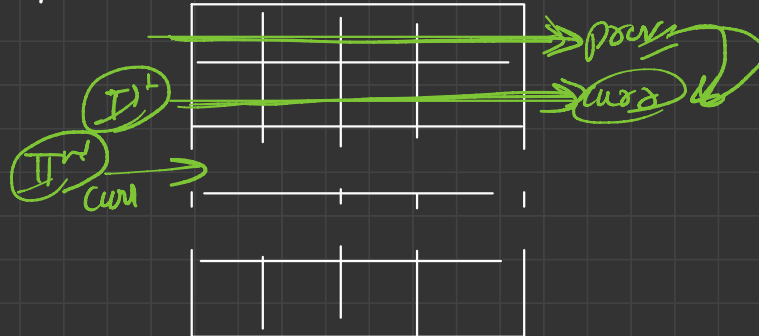
0 $\rightarrow \maxWgt$

1 $\rightarrow n-1$

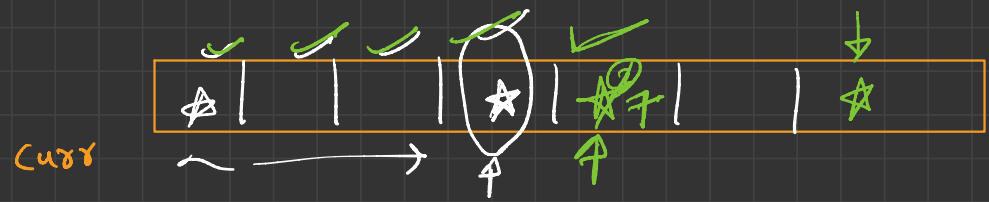
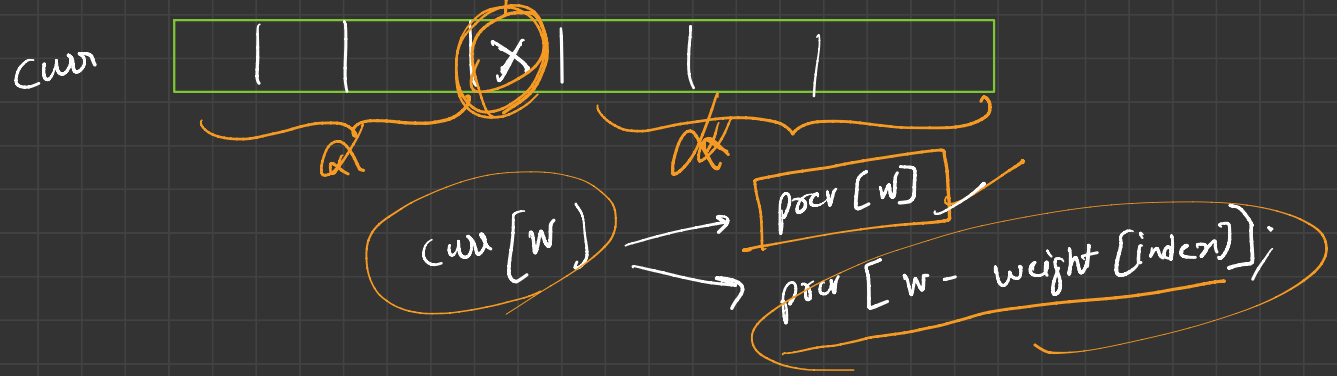
for $(=0 \leq \maxWeight + 1)$

for $(=1, < n)$

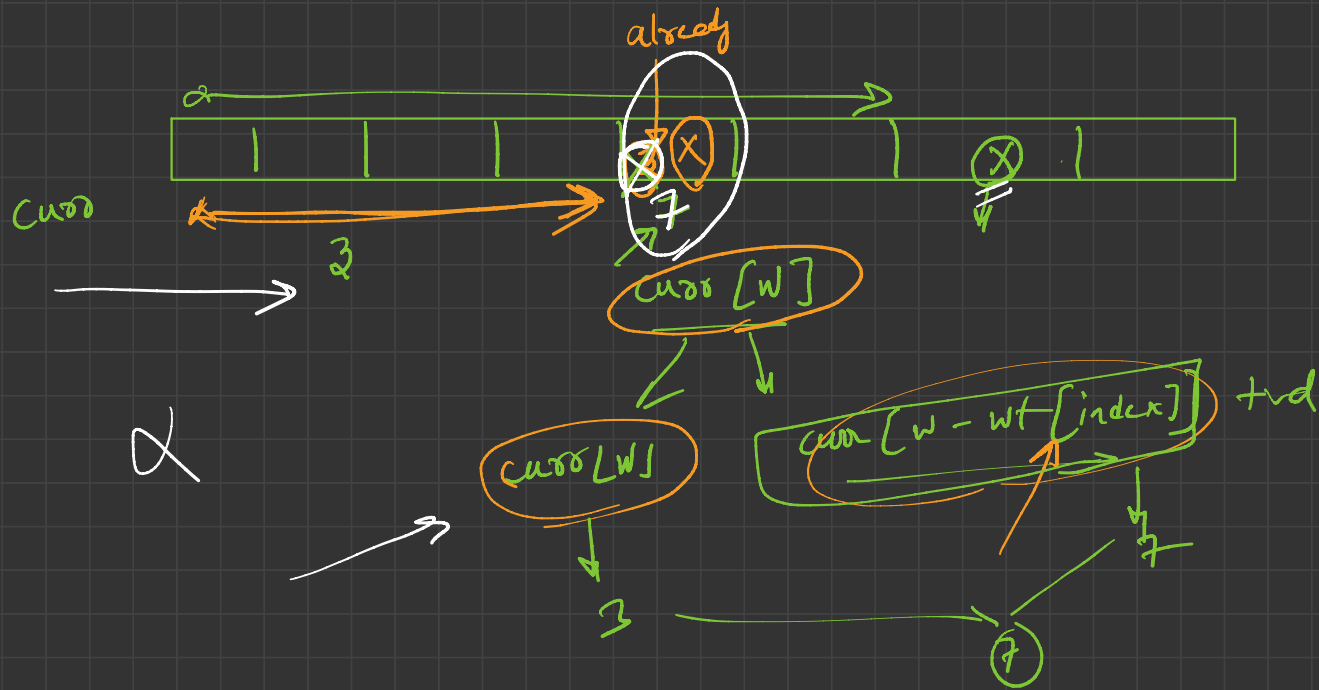
S.O $\rightarrow$



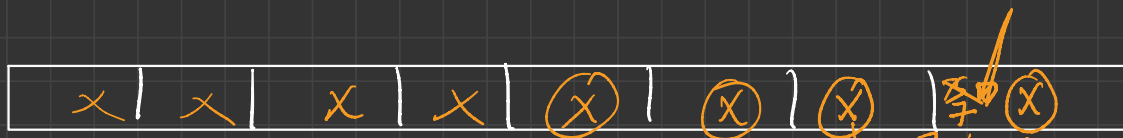
or can be improve?



$exclude \rightarrow cur[w]$
 $ind \rightarrow \text{---} + cur[w - weight[ind]]$



curr



Recursive →

+ memoisation //

$O(n \times W)$ → S.C

Tabulation → $O(n \times W)$

S.O → 0

S.C → $O(2 \times W)$

$O(n)$

